

Lab 1: Requirements Engineering & UML Use-Case Modelling

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Problem Statement #23

System Overview

- The Hyperlocal Courier Dispatch & Tracking Engine is an on-demand package delivery platform for local courier services.
- It allows customers to book parcel pickups and deliveries through the system.
- The system automatically assigns the nearest available delivery rider to reduce pickup and delivery time.
- It is designed to provide a fast, efficient, and secure courier service.
- The platform is mainly used by Sender Clients, Delivery Riders, and Administrators.
- It enables live GPS tracking, allowing customers to monitor their parcel in real time.
- The system supports multi-stop route optimization to help riders complete multiple deliveries efficiently.
- It uses OTP verification to ensure that parcels are delivered only to the intended recipient.
- The system also provides delivery status updates and notifications throughout the delivery process.
- Overall, it improves delivery efficiency, enhances customer satisfaction, and ensures secure parcel delivery.

In Scope

The Hyperlocal Courier Dispatch & Tracking Engine supports the following functionalities:

- Allows users to book courier pickup and delivery requests.

- Automatically assigns the nearest available delivery rider.
- Provides real-time GPS tracking of the delivery.
- Verifies successful parcel delivery using OTP authentication.
- Updates the delivery status from pickup to successful completion.
- Sends delivery notifications and tracking updates to the sender.

Out of Scope

The following features are not included in the current system:

- International or cross-border courier services.
- Warehouse and inventory management.
- Delivery rider recruitment and fleet maintenance.
- Customer support and complaint management.
- Processing returns or reverse logistics.
- Integration with third-party logistics providers.

Actor	Role	Description
Sender Client	Primary Actor	Creates courier requests, enters pickup and delivery details, tracks parcel status in real time, and receives delivery notifications.
Delivery Rider	Primary Actor	Accepts courier assignments, picks up parcels, updates delivery status, follows the optimized route, and completes delivery after OTP verification.
OTP Service	Supporting Actor	Generates and validates One-Time Passwords (OTPs) to ensure secure parcel handover to the intended recipient.
Admin	Supporting Actor	Monitors system operations, manages users and delivery activities, and oversees the overall performance and reliability of the platform.

Functional Requirements

FR-001

Description:

The system shall match an outgoing courier request with the nearest active delivery rider within a 3 km radius.

Priority:

High

Acceptance Criteria:

- Pass: The rider receives a dispatch notification with the pickup location.
- Fail: The order is assigned to an offline or unavailable rider.

Rationale:

Minimizes pickup waiting time and ensures that only available riders are assigned to delivery requests.

FR-002**Description:**

The system shall allow the Sender Client to create a courier request by entering the pickup address, drop-off address, package details, and preferred pickup time.

Priority:

High

Acceptance Criteria:

- Pass: The request is successfully created, and a tracking ID is generated when all mandatory details are valid.
- Fail: The request is rejected if mandatory information is missing or invalid.

Rationale:

Ensures that all essential information is collected before assigning a delivery rider.

FR-003**Description:**

The system shall generate an optimized multi-stop route whenever a delivery rider is assigned multiple pickups or deliveries in a single trip.

Priority:

Medium

Acceptance Criteria:

- Pass: The generated route minimizes the total travel distance and delivery time.
- Fail: Stops are displayed without route optimization.

Rationale:

Reduces travel time, fuel consumption, and improves delivery efficiency during peak hours.

FR-004**Description:**

The system shall require the Delivery Rider to verify the recipient's One-Time Password (OTP) before marking the parcel as delivered.

Priority:

High

Acceptance Criteria:

- Pass: The delivery status changes to "Delivered" only after a valid OTP is entered.
- Fail: The parcel is marked as delivered without OTP verification or with an incorrect OTP.

Rationale:

Ensures secure parcel delivery and prevents unauthorized delivery confirmation.

FR-005**Description:**

The system shall allow the Sender Client to view the live location of the assigned delivery rider along with the current delivery status.

Priority:

Medium

Acceptance Criteria:

- Pass: The rider's location and delivery status are updated automatically in real time.
- Fail: The location or delivery status does not update after the rider's status changes.

Rationale:

Provides transparency, improves customer trust, and reduces the need for customer support inquiries.

Non-Functional Requirements

NFR-001**Description:**

The system shall transmit delivery status and live GPS telemetry updates from the delivery rider to the Sender Client with a latency of less than 2 seconds.

Priority:

High

Acceptance Criteria:

- Pass: Performance testing confirms that delivery status and GPS updates are transmitted within 2 seconds under simulated peak load.
- Fail: Updates exceed the 2-second latency requirement or fail to meet the specified performance standard.

Rationale:

Provides near real-time tracking, improves user experience, and increases customer trust in the delivery platform

NFR-002**Description:**

The dispatch matching engine shall maintain 99.9% system availability and assign new courier requests within 5 seconds while handling up to 5,000 concurrent active requests.

Priority:

High

Acceptance Criteria:

- Pass: Load testing demonstrates system uptime of at least 99.9% and rider matching within 5 seconds under 5,000 concurrent requests.
- Fail: System availability falls below 99.9% or rider assignment exceeds 5 seconds during testing.

Rationale:

Ensures the platform remains reliable, scalable, and capable of handling high demand during peak operational hours.

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match an outgoing courier request with the nearest active delivery rider within a 3 km radius.	High	Pass: Rider receives a dispatch notification with the pickup location. Fail: Order is assigned to an offline or unavailable rider.	Minimizes pickup waiting time and ensures only available riders are assigned.
FR-002	Functional	The system shall allow the Sender Client to create a courier request by entering the pickup address, drop-off address, package details, and preferred pickup time.	High	Pass: Request is created successfully and a tracking ID is generated. Fail: Request is rejected if mandatory details are missing or invalid.	Ensures all required information is collected before rider assignment.
FR-003	Functional	The system shall generate an optimized multi-stop route whenever a rider has multiple deliveries in a single trip.	Medium	Pass: The optimized route reduces travel distance and delivery time. Fail: Stops are displayed without optimization.	Improves delivery efficiency and reduces fuel consumption.

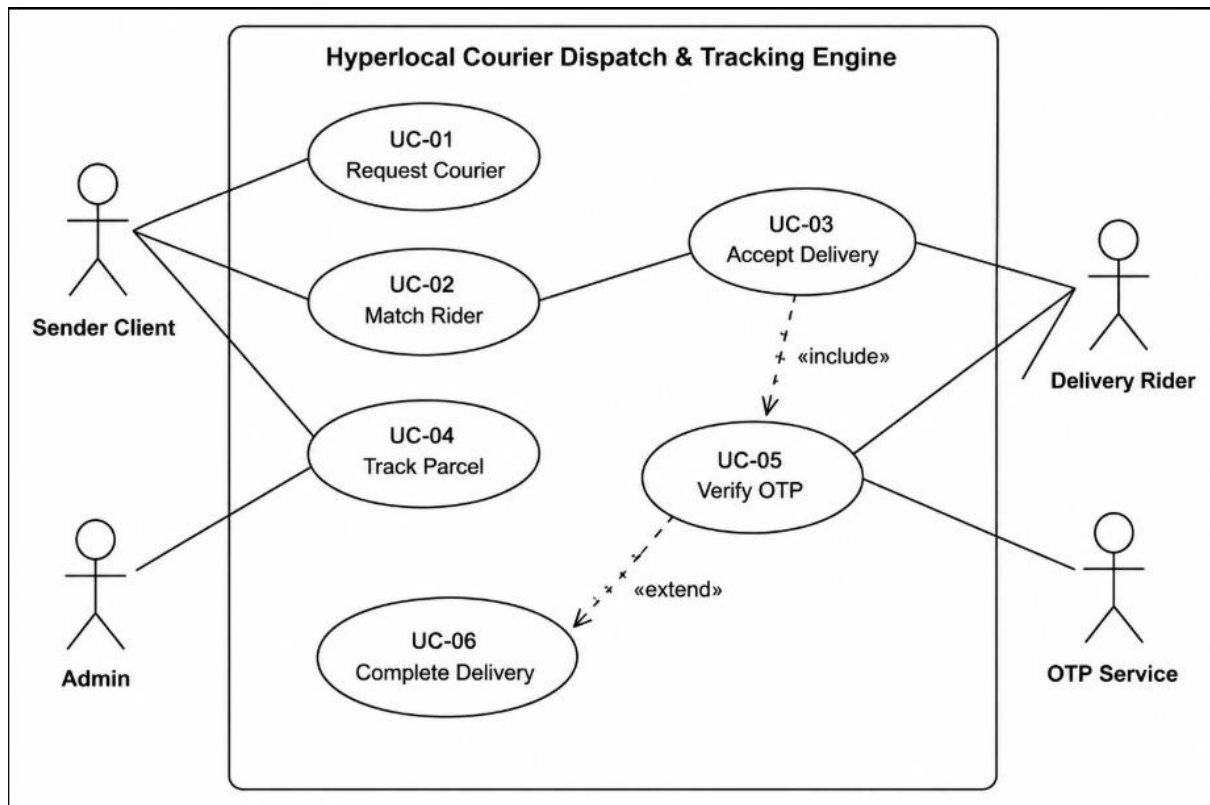
FR-004	Functional	The system shall require OTP verification before marking a parcel as delivered.	High	Pass: Delivery is completed only after entering the correct OTP. Fail: Delivery is marked complete without OTP or with an incorrect OTP.	Ensures secure delivery and prevents unauthorized parcel handover.
FR-005	Functional	The system shall allow the Sender Client to view the live location of the assigned rider and the current delivery status.	Medium	Pass: Live location and delivery status update automatically. Fail: Location or status does not refresh after rider movement or status change.	Improves transparency, customer confidence, and tracking accuracy
NFR-001	Non-Functional	The system shall transmit delivery status and live GPS updates with a latency of less than 2 seconds.	High	Pass: Performance tests confirm updates are delivered within 2 seconds under peak load. Fail: Updates exceed the specified latency.	Provides near real-time tracking and enhances user experience.
NFR-002	Non-Functional	The dispatch matching engine shall maintain 99.9% uptime and assign new requests within 5 seconds while handling up to 5,000 concurrent requests.	High	Pass: Load testing confirms 99.9% uptime and rider matching within 5 seconds. Fail: System availability or matching performance falls below the	Ensures platform reliability, scalability, and consistent performance during peak demand.

				specified requirements.	
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Identified Use Cases

UC ID	Use Case	Description
UC-01	Request Courier	Allows the Sender Client to create a new courier request by entering pickup, delivery, and parcel details.
UC-02	Match Rider	Automatically identifies and assigns the nearest available delivery rider to the courier request.
UC-03	Accept Delivery Request	Enables the Delivery Rider to accept or reject the assigned courier request.
UC-04	Track Parcel	Allows the Sender Client to view the live location and current delivery status of the parcel.
UC-05	Verify OTP	Validates the One-Time Password entered by the receiver before completing the delivery
UC-06	Complete Delivery	Marks the parcel as successfully delivered after successful OTP verification and updates the delivery status.

UML Use-Case Diagram



11. Use-Case Specification

Use Case ID

UC-06

Use Case Name

Complete Delivery

Primary Actor

Delivery Rider

Secondary Actor

Sender Client, OTP Service

Preconditions

- Parcel has been picked up.
- Rider has reached the delivery location.
- Receiver has received the OTP.

Trigger

The rider reaches the delivery location.

Postconditions

- OTP is verified.
- Parcel is marked as delivered.
- Sender receives a delivery confirmation.

Main Success Scenario

1. Rider reaches the delivery location.
2. Rider requests the OTP from the receiver.
3. Receiver provides the OTP.
4. System verifies the OTP.
5. Parcel is handed over to the receiver.
6. System marks the delivery as completed.
7. Sender receives a confirmation notification.

Alternate Flow

1. Receiver enters an incorrect OTP.
2. System displays an error message.
3. Receiver re-enters the OTP.
4. If the OTP is valid, the delivery continues.
5. If the OTP remains invalid after multiple attempts, the delivery is not completed and the issue is reported.

